

ECON 0100 | Exercise B4 | Price Controls

I Solutions

Answers and work are shown in the red blocks.

Pumpkin pasties again, with the same demand and supply curves as Exercise B3:

$$P = 12 - \frac{Q_d}{2} \qquad P = 2 + \frac{Q_s}{2}$$

Prices are in galleons and quantity is in pasties. The equilibrium you found is 10 pasties at 7 galleons.

First, the market before any control. $Q^* = 10$ at $P^* = 7$, so

$$CS = (12 - 7) \cdot 10 \cdot \frac{1}{2} = 25 \qquad PS = (7 - 2) \cdot 10 \cdot \frac{1}{2} = 25$$

Total surplus is 50. Each control below is measured against this.

Q1 | A Price Ceiling

The government sets a maximum legal price of 5 galleons. From Exercise B3, at 5 galleons the quantity demanded is 14 and the quantity supplied is 6.

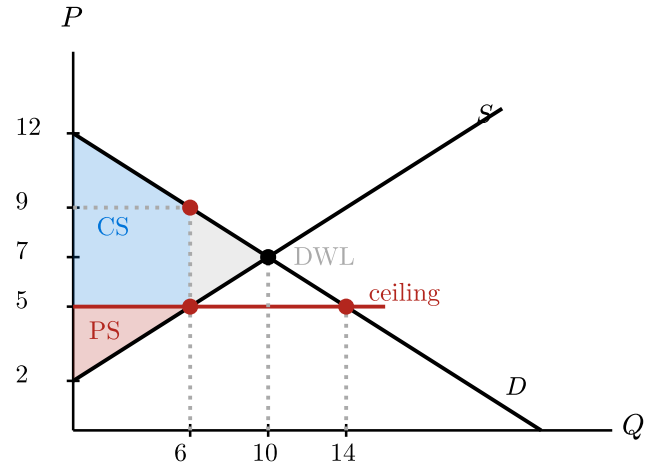
Start by plotting the demand curve, the supply curve, and the price ceiling, and shade consumer surplus, producer surplus, and deadweight loss.

- a) How many pasties are exchanged? _____
- b) What is consumer surplus? _____
- c) What is producer surplus? _____
- d) What is deadweight loss? _____

I a) 6 pasties. b) $CS = 33$ galleons. c) $PS = 9$ galleons. d) $DWL = 8$ galleons.

a) The ceiling is below P^* , so it binds. Buyers want 14 but sellers only bake 6 at 5 galleons: a shortage of 8, and only the short side's 6 pasties are exchanged.

b) Consumer surplus is everything below demand and above the price out to $Q = 6$, a triangle on top of a rectangle. We need to find V , the height of demand at 6: $V = 12 - \frac{1}{2} \cdot 6 = 9$.



$$CS = (12 - 9) \cdot 6 \cdot \frac{1}{2} + (9 - 5) \cdot 6 = 9 + 24 = 33$$

c) Producer surplus is the triangle above supply and below the price out to $Q = 6$:

$$PS = (5 - 2) \cdot 6 \cdot \frac{1}{2} = 9$$

d) Deadweight loss is the triangle between the curves from 6 to 10, the surplus on the 4 pasties no longer traded:

$$DWL = (9 - 5) \cdot (10 - 6) \cdot \frac{1}{2} = 8$$

Buyers gain 8 (25 to 33), sellers lose 16 (25 to 9), and the market loses 8: $33 + 9 + 8 = 50$.

Q2 | A Price Floor

Suppose instead the government sets a minimum legal price of 9 galleons. Now the quantity demanded is 6 and the quantity supplied is 14.

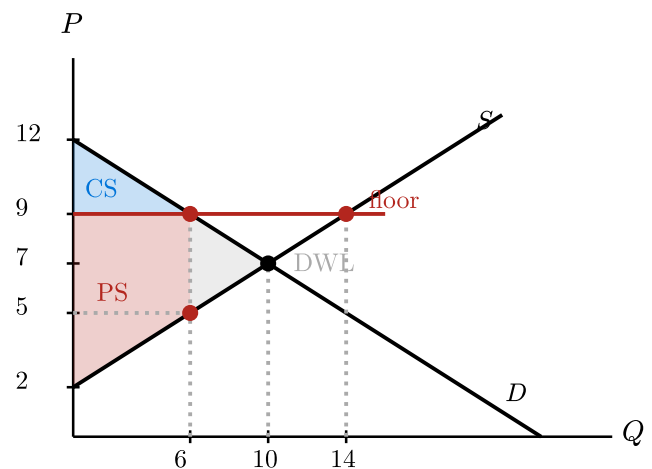
Start by plotting the demand curve, the supply curve, and the price floor, and shade consumer surplus, producer surplus, and deadweight loss.

- How many pasties are exchanged? _____
- What is producer surplus? _____
- What is deadweight loss? _____
- Would a floor of 6 galleons change the market? _____

a) 6 pasties. b) PS = 33 galleons. c) DWL = 8 galleons. d) No.

a) The floor is above P^* , so it binds. Sellers bring 14 but buyers only take 6 at 9 galleons: an excess of 8, and again the short side's 6 pasties are exchanged.

b) Now the sellers hold the rectangle. V is the height of supply at 6: $V = 2 + \frac{1}{2} \cdot 6 = 5$.



$$PS = (9 - 5) \cdot 6 + (5 - 2) \cdot 6 \cdot \frac{1}{2} = 24 + 9 = 33$$

Consumer surplus is the small triangle, $(12 - 9) \cdot 6 \cdot \frac{1}{2} = 9$.

c) The same 4 lost pasties as the ceiling, so the same triangle:

$$DWL = (9 - 5) \cdot (10 - 6) \cdot \frac{1}{2} = 8$$

The floor is the ceiling's mirror image: sellers gain 8, buyers lose 16, the market loses 8.

d) No. A floor of 6 sits below the equilibrium price of 7, so the market clears above it and the floor never binds: still 10 pasties at 7 galleons, with no excess and no deadweight loss. A control only changes the market when it blocks the equilibrium price.